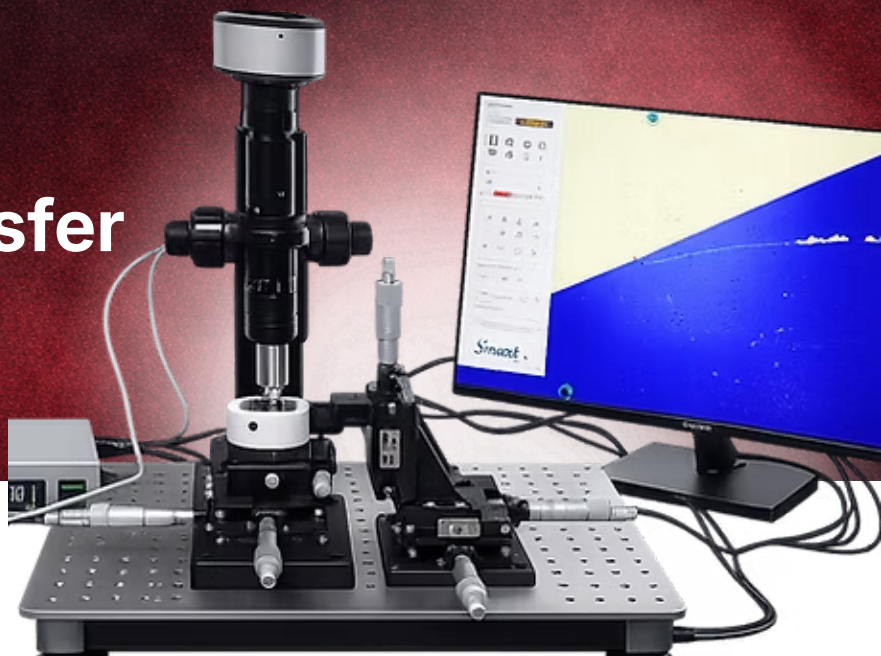


2DTS-M2 Manual Deterministic Transfer System

Manual / semi-manual PDMS-based 2D transfer platform



Accurate Alignment.
Controlled Pickup & Release.
Built for Research-Grade 2D Material Assembly.

The CryoNano 2DTS-M2 is a high-precision motorized deterministic transfer platform engineered for advanced assembly of 2D materials, van der Waals heterostructures, and hybrid nanoscale devices.

Designed for research laboratories and semiconductor prototyping environments, the system enables accurate flake alignment, controlled transfer, and repeatable multi-layer stacking with sub-micron precision. The integrated software-controlled architecture provides synchronized motion control, optical imaging, temperature-assisted transfer, and automated positioning for deterministic assembly workflows.

KEY FEATURES

- **Deterministic Dry Transfer**
PDMS-based viscoelastic stamping enables controlled pickup and release of 2D flakes with minimal contamination and excellent repeatability.
- **Precision Motion Control**
High-resolution XYZ and XYθ stages provide accurate alignment, contact control, and peel-off manipulation during transfer.
- **Real-Time Optical Alignment**
Integrated epi-illumination microscope with HDMI camera output enables live flake visualization and precise positioning.
- **Modular System Architecture**
Mechanical, optical, and thermal modules can be configured for a wide range of experimental workflows and research environments.
- **Optional Temperature-Controlled Transfer**
Integrated heater and PID temperature controller support temperature-assisted transfer processes and enhanced adhesion control.

Notes

- Transfer performance depends on vibration isolation and environmental stability
- Fine control of peel-off rate is critical for successful flake release
- Compatible with diverse substrates including Si/SiO₂ wafers
- Optional thermal module available for heated transfer workflows
- Minor PDMS residue may remain after transfer and is typically optically transparent and chemically inert
- Raman or photoluminescence spectroscopy may be used for thickness verification

APPLICATIONS

- **Deterministic transfer of 2D materials**
- **Twistronics and moiré device assembly**
- **Graphene and TMDC heterostructure research**
- **Quantum device prototyping**
- **Semiconductor and nanoelectronics research**
- **Air-sensitive material transfer inside glove-box environments**
- **Advanced materials characterization workflows**



2DTS-M2 Manual Deterministic Transfer System

SPECIFICATIONS

LINEAR MOTION SYSTEM

Stage Type	Linear crossed-roller stage
Material	Anodized aluminum / stainless steel
Travel Range	>25 mm
Guiding Mechanism	Crossed-roller bearings
Motion Type	Motorized linear motion

THETA ROTATION STAGE

Stage Type	Rotary theta stage
Drive Mechanism	Worm gear transmission
Bearing Type	Crossed-roller
Chuck Material	OFHC copper vacuum chuck
Angular Resolution	0.1° to 0.01°
Angular Accuracy	±0.02°
Flatness	≤ ±5 μm

TEMPERATURE CONTROL

Temperature Range	RT to 100°C
Control Method	PID
Resolution	0.1°C
Uniformity	Full-stage uniform heating

INTEGRATED IMAGING & MEASUREMENT

Measurement Tools	Distance & angle measurement
Overlay Tools	Adjustable line overlays
Video Functions	Freeze / playback
Capture Functions	Image & video export

PRECISION PERFORMANCE

Position Accuracy	±1 μm to ±5 μm
Straightness & Flatness	≤ ±2 μm
Repeatability	High precision closed-loop control

MOTOR CONTROLLER

Axis Support	Single-axis / multi-axis
Interface	USB + digital I/O
Software	GUI with scripting support
Input Power	24 V DC

OPTICAL SYSTEM

Optical Type:	Apochromatic motorized system
Zoom Range:	0.3X–3.0X
Zoom Ratio:	10:1
Objective Type:	20X infinity corrected
FOV Enhancement:	Up to 1.5X larger than conventional systems
Sensor:	Sony CMOS